EXP NO-02

write a C program to simulate a NoN Deterministic finite AUtomata(NFA) for the given language representing strings that start with 0 and end with 1

C program

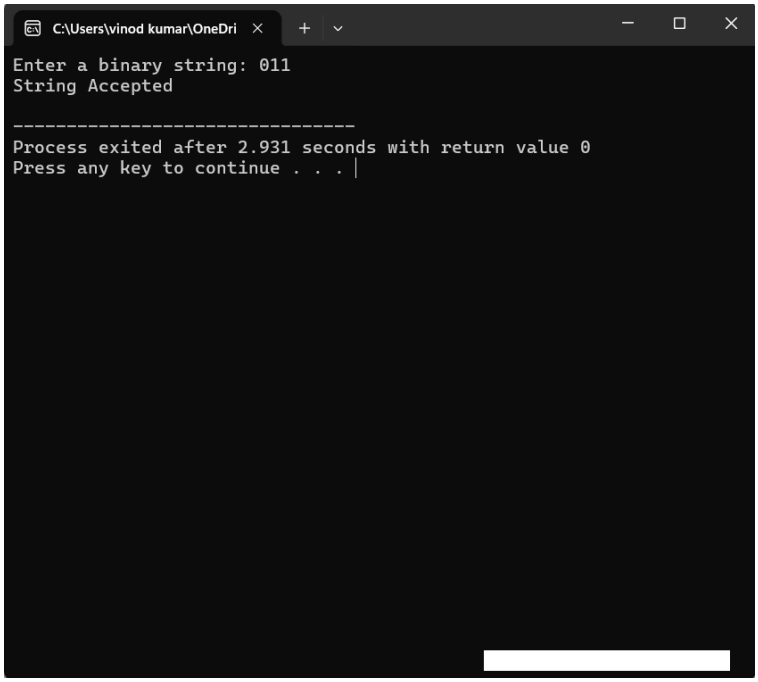
```
#include <stdio.h>
#include <string.h>
int main() {
  char str[100];

  // States: q0 (start), q1, q2 (final), qd (dead)
  int current[4] = {1, 0, 0, 0};
  int next[4];
  printf("Enter a binary string: ");
  scanf("%s", str);
  for(int i = 0; i < strlen(str); i++) {
    next[0] = next[1] = next[2] = next[3] = 0;
    if(current[0]) { // q0
      if(str[i] == '0')
        next[1] = 1;
      else
        next[3] = 1;
    }
    if(current[1]) { // q1
      if(str[i] == '0')
        next[1] = 1;
      if(str[i] == '1') {
        next[1] = 1;
        next[2] = 1;
```

```
        }
      }
      if(current[2]) { // q2 (final state)
        if(str[i] == '0')
          next[1] = 1;
        if(str[i] == '1') {
          next[1] = 1;
          next[2] = 1;
        }
      }
      if(current[3]) { // dead state
        next[3] = 1;
      }
      for(int j = 0; j < 4; j++)
        current[j] = next[j];
    }
    if(current[2])
      printf("String Accepted\n");
    else
      printf("String Rejected\n");
    return 0;
}
```

INPUT1:

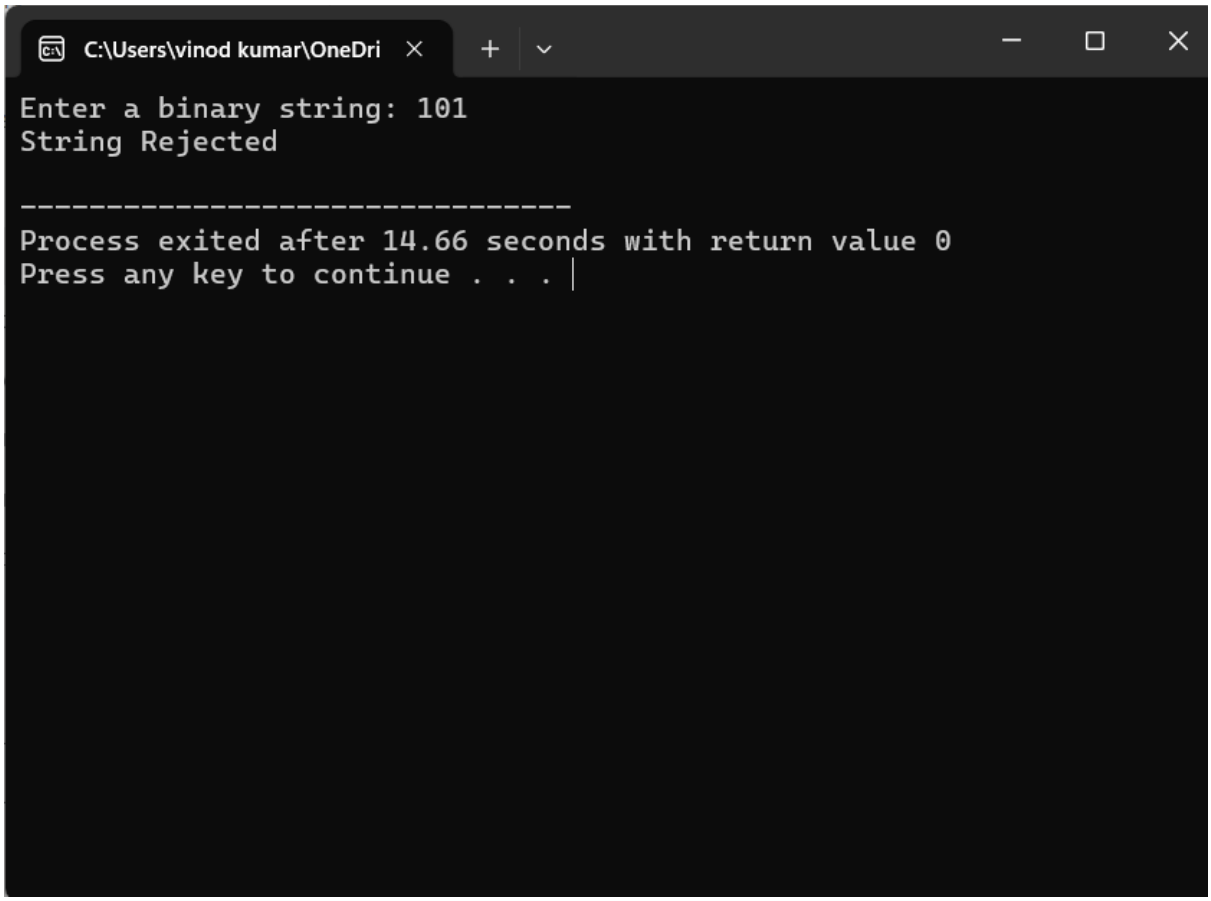
011


C:\Users\vinod kumar\OneDri
Enter a binary string: 011
String Accepted

Process exited after 2.931 seconds with return value 0
Press any key to continue . . .

INPUT2:

101


C:\Users\vinod kumar\OneDri
Enter a binary string: 101
String Rejected

Process exited after 14.66 seconds with return value 0
Press any key to continue . . .